

Run 07 | TEA

Global (including India)

IV + controls

Target Total early-stage entrepreneurial activity; binary 1 vs. 0

Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022

Positive cases 154,420 (11.50%) **Case time** 31m 57s

Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)

Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9934). Highest F1: Random Forest (0.8835). For XGBoost, adding controls changed ROC-AUC by +0.0052 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9934	0.9501	97.29	87.81	88.77	0.8829	15s
Random Forest	0.9934	0.9501	97.26	86.46	90.33	0.8835	1m 16s
CatBoost	0.9931	0.9475	97.17	86.12	89.92	0.8798	1m 50s
H2O GBM	0.9919	0.9356	96.89	84.88	88.80	0.8679	20s
Lasso	0.9909	0.9289	96.79	84.59	88.11	0.8632	8m 45s
Elastic Net	0.9909	0.9289	96.79	84.58	88.12	0.8632	7m 35s
Logistic Regression	0.9907	0.9285	96.78	84.28	88.54	0.8636	1m 32s
Ridge	0.9906	0.9267	96.73	84.42	87.74	0.8605	7m 12s
Decision Tree	0.9889	0.8920	96.67	84.45	87.09	0.8575	6s
Naive Bayes	0.9390	0.5855	91.57	59.56	83.29	0.6945	2m 19s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	74,143.72	47.24
2	Established business owner	30,763.67	19.60
3	Occupational status	12,422.00	7.91
4	GDP per capita (PPP)	4,556.58	2.90
5	Population	4,146.25	2.64
6	Age	3,926.84	2.50
7	Self-efficacy	3,912.17	2.49
8	Knowing entrepreneurs	3,443.57	2.19

Share is the percentage of total forest impurity importance across all 40 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Source: Completed Objective 1 batch, 27 September 2026; 07_obj1_global_iv_cv_TEA. Scores and importance are taken directly from saved CSV outputs.